

Astronomical Bycatch

Proof of concept for AI-enabled exoplanet transit detection in archival data

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Abstract

Astronomical Bycatch is an ambitious project to extract exoplanet transit signals from archival data by leveraging recent advances in Artificial Intelligence. Observatories around the world have made thousands of digitally recorded observations every night for decades, creating archives with petabytes of data. In almost every astronomical observation, no matter the intended target, “background” stars are captured as “Bycatch.”

Until recently, reprocessing and procedurally searching through such archives had been an intractable task. Recent advances in AI-enabled time-series data analysis have made it possible, but challenging. Though the probability that any given observation reveals a novel transit is low, their sheer number means that we expect to detect many transiting exoplanets. Here, we present the results of feasibility and scalability studies carried out at DCU using archival data of known transits.

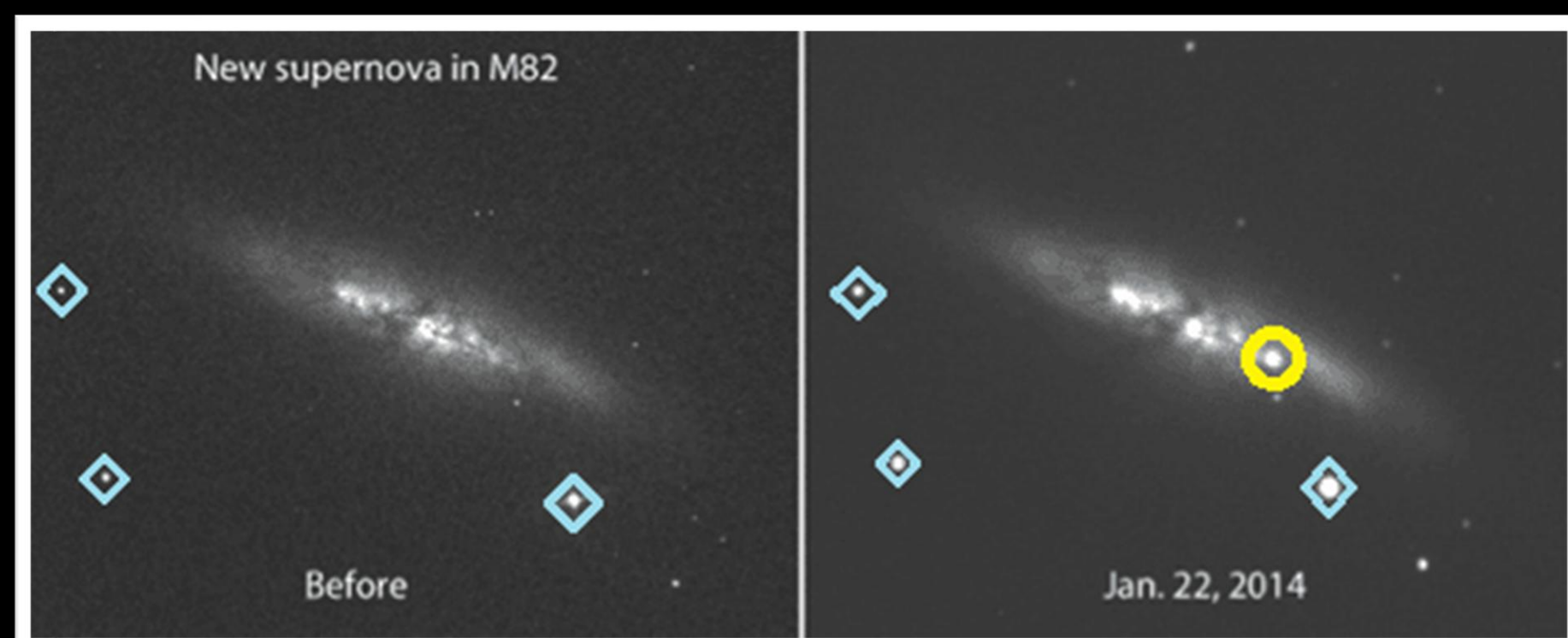
Concept

Many observatories around the world

Many observations each night

Many frames per observation

Many “Bycatch” stars in each frame



Stars (in blue diamonds) are an example of Astronomical Bycatch in the foreground of these images of M82 that Guido et al. captured to observe the supernova (yellow circle).

Decades of imaging data recorded in archives

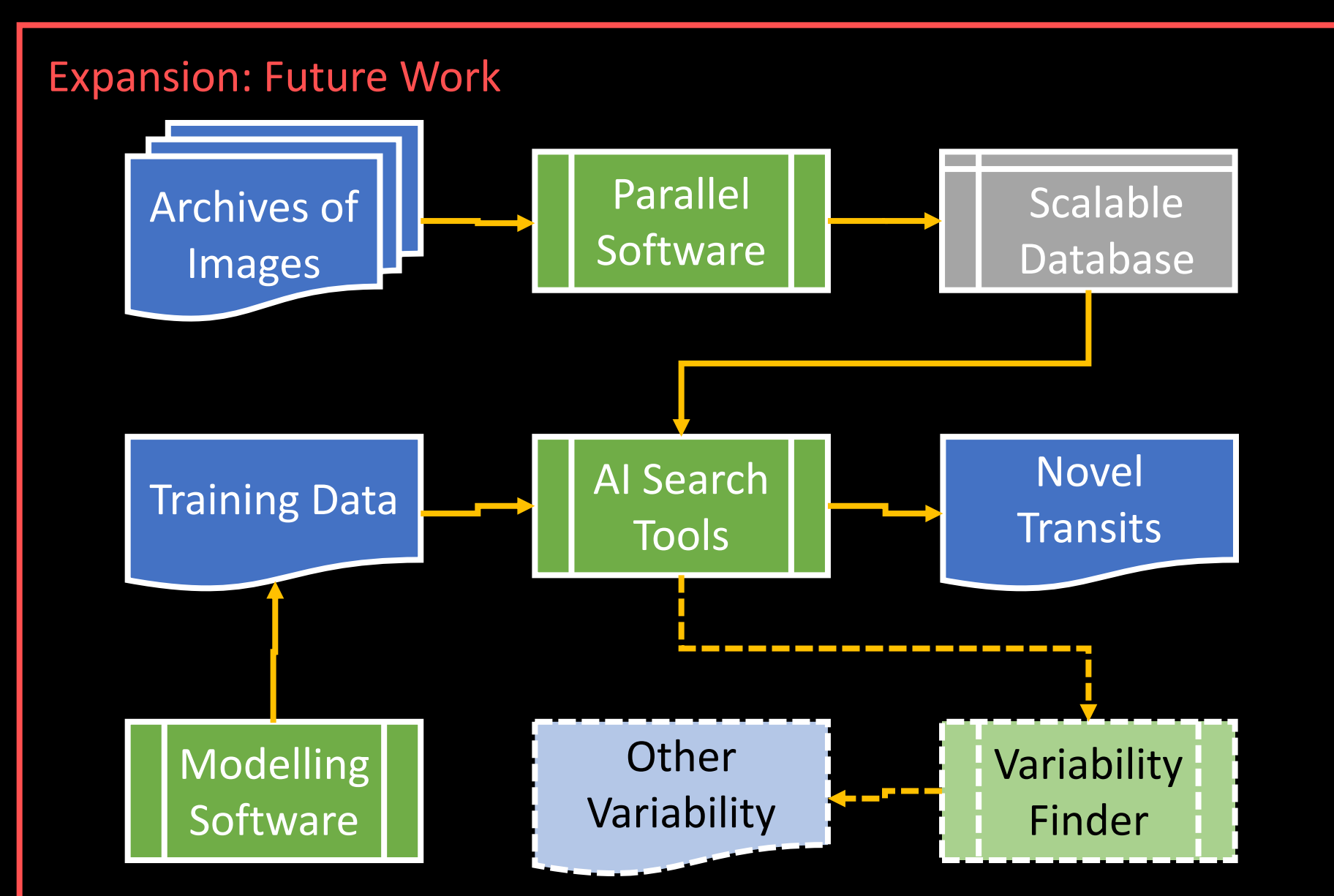
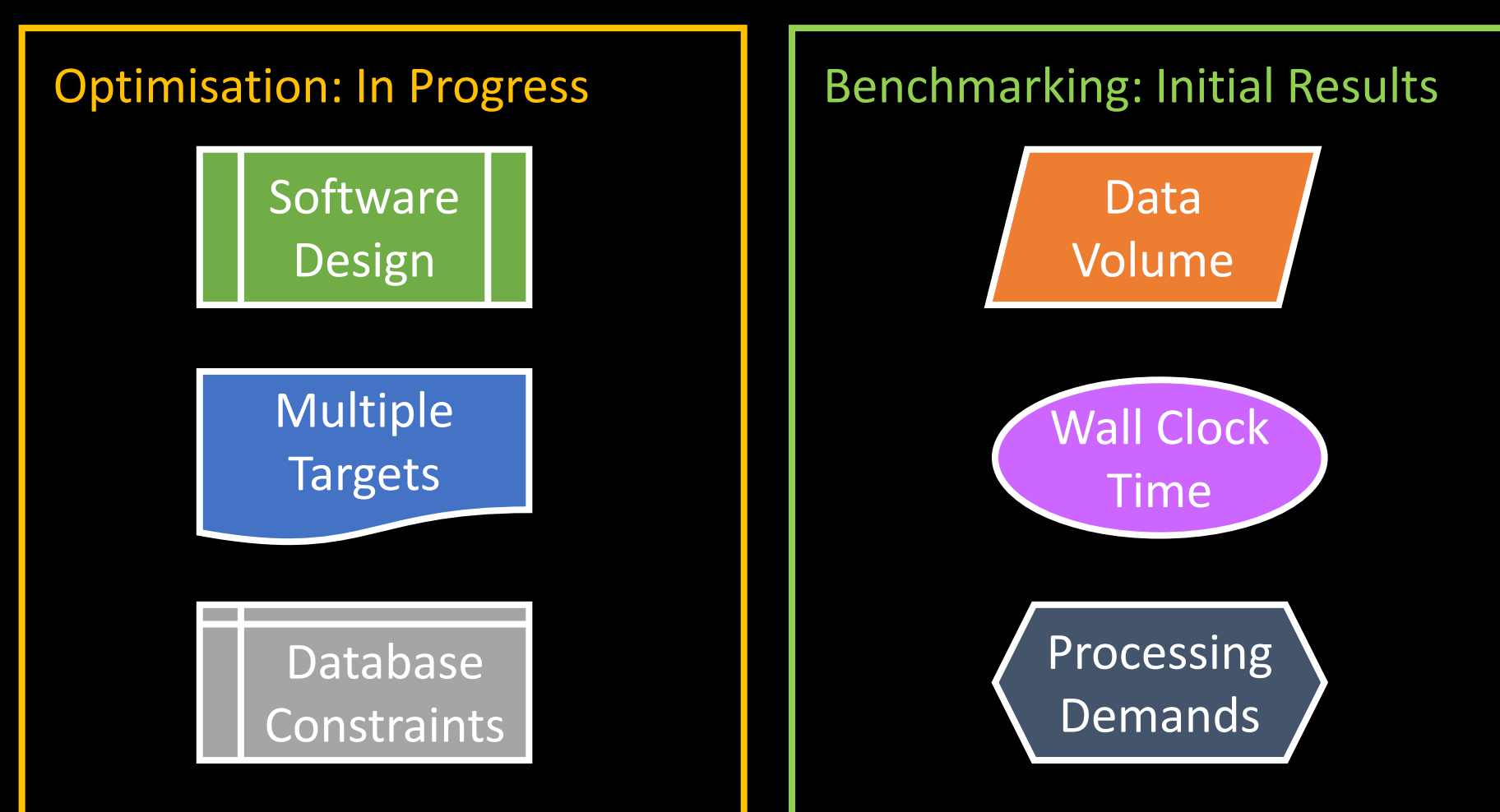
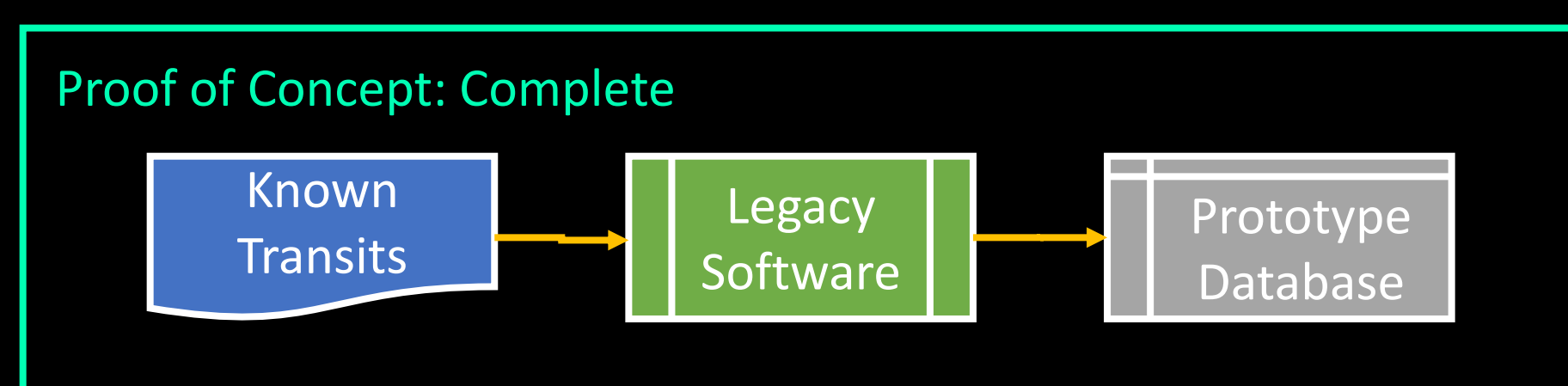
Petabytes of images (e.g. ESO, ESA Archives)

Modest (~1%) photometric precision required

Serendipitous observation of transits must occur

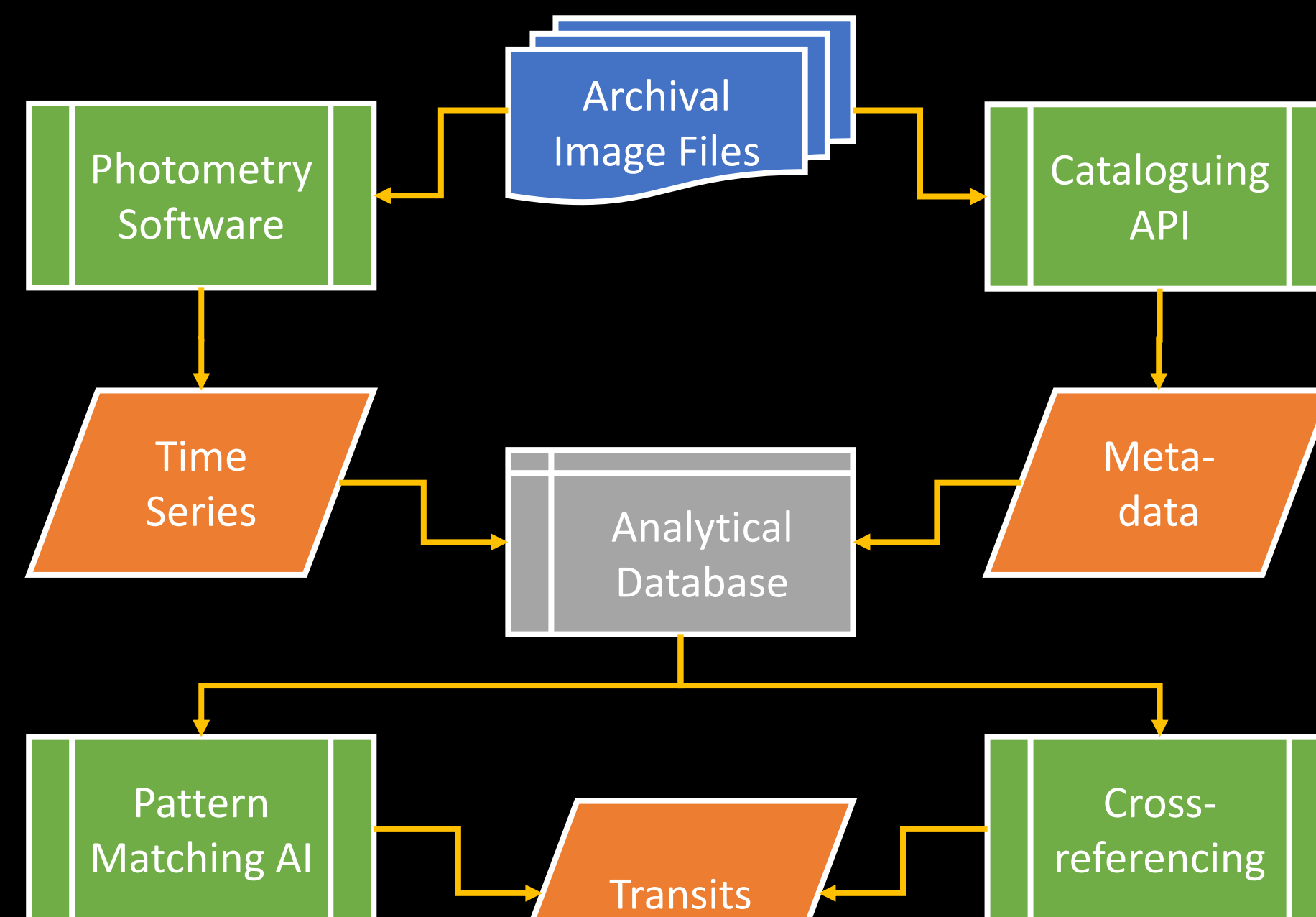
Challenge: trawling through the data to find them!

Plan and Progress



High-level plan for the stages of the project.

Workflow



High-level workflow of the overall system.

Images processed into photometric time series

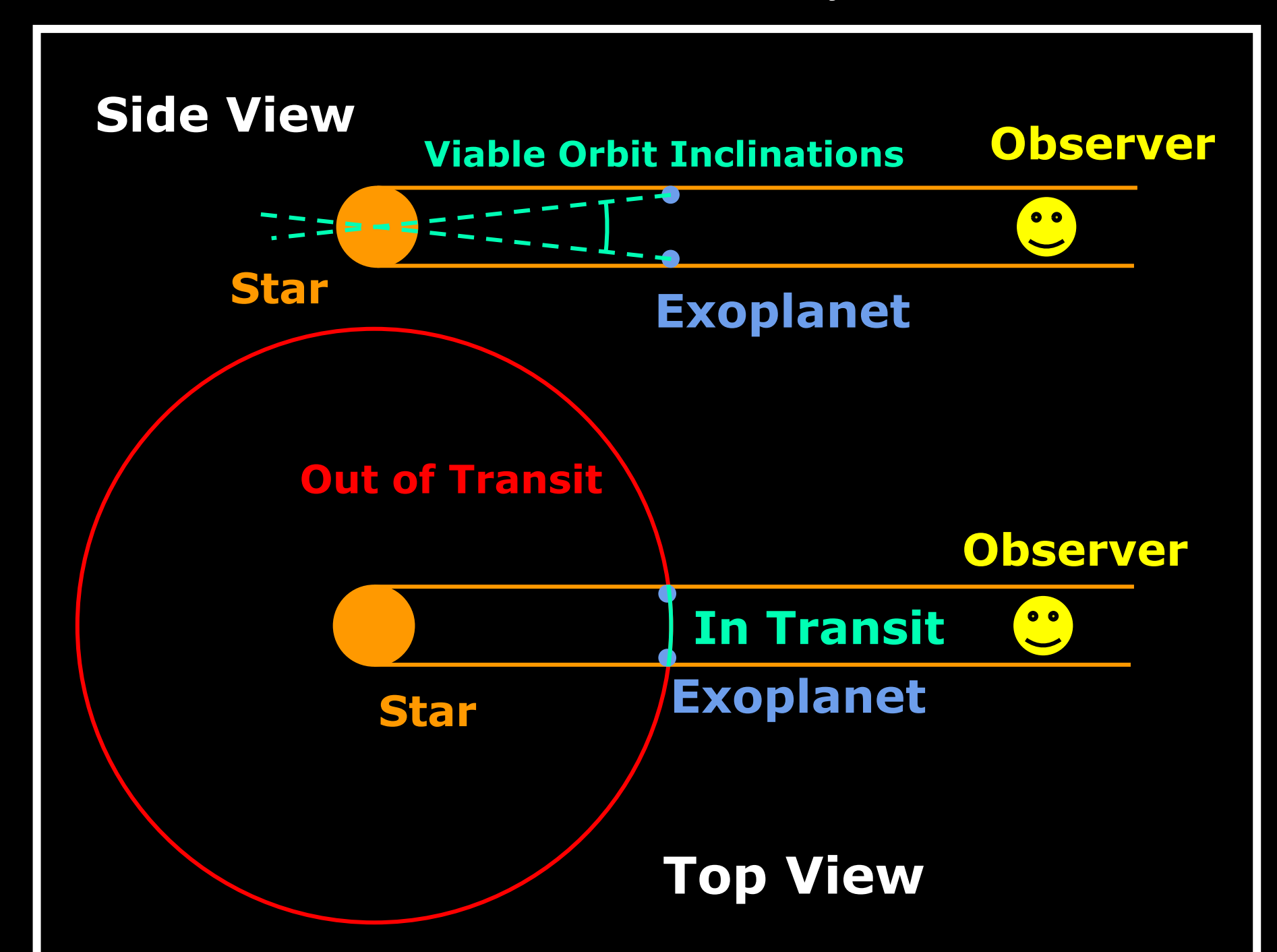
Stored in a database with Metadata

Heterogenous data sources collated

AI Pattern-matching used to extract transits

Follow-up observations to confirm results

Geometry



An illustration of the constraints on orbital alignment for an exoplanet transit. The top image shows that transits can only be observed for a relatively small proportion of orbital alignments, while the bottom image shows that transits durations are short relative to orbital period.

Probability of orbital alignment

Low: Earth $\approx 0.47\%$

Proportion of time in transit

Low: Earth ≈ 13 hours/year

Mean number of observations per transit

High: Earth $\approx 143,000$

Proof of concept & benchmarks

Demonstrated concept using known targets, legacy software and prototype database

Developed data extraction and ingestion tools

Benchmarked legacy software and novel tools

Identified photometric data extraction as the dominant bottleneck in wall-clock time

Identified optimisations:

- Process multiple targets at once
- Remove intermediate file I/O steps

Next Steps

Demonstrate detection of historical transits of known exoplanet hosts

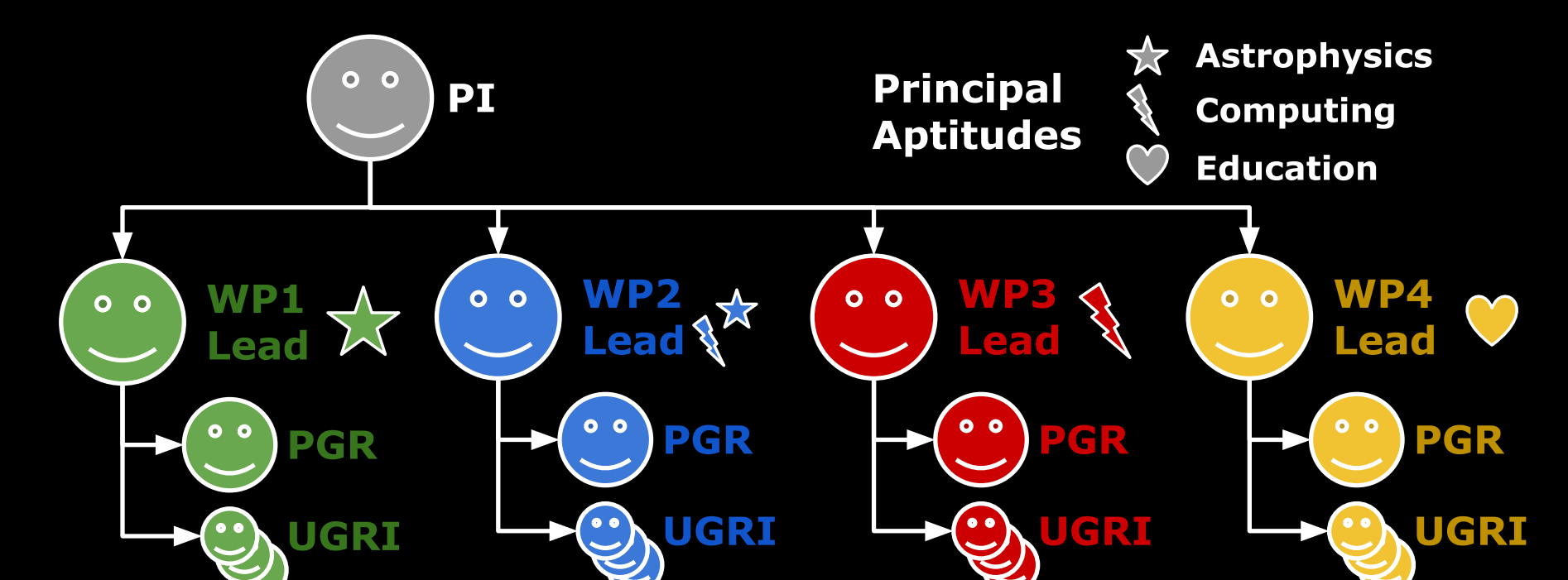
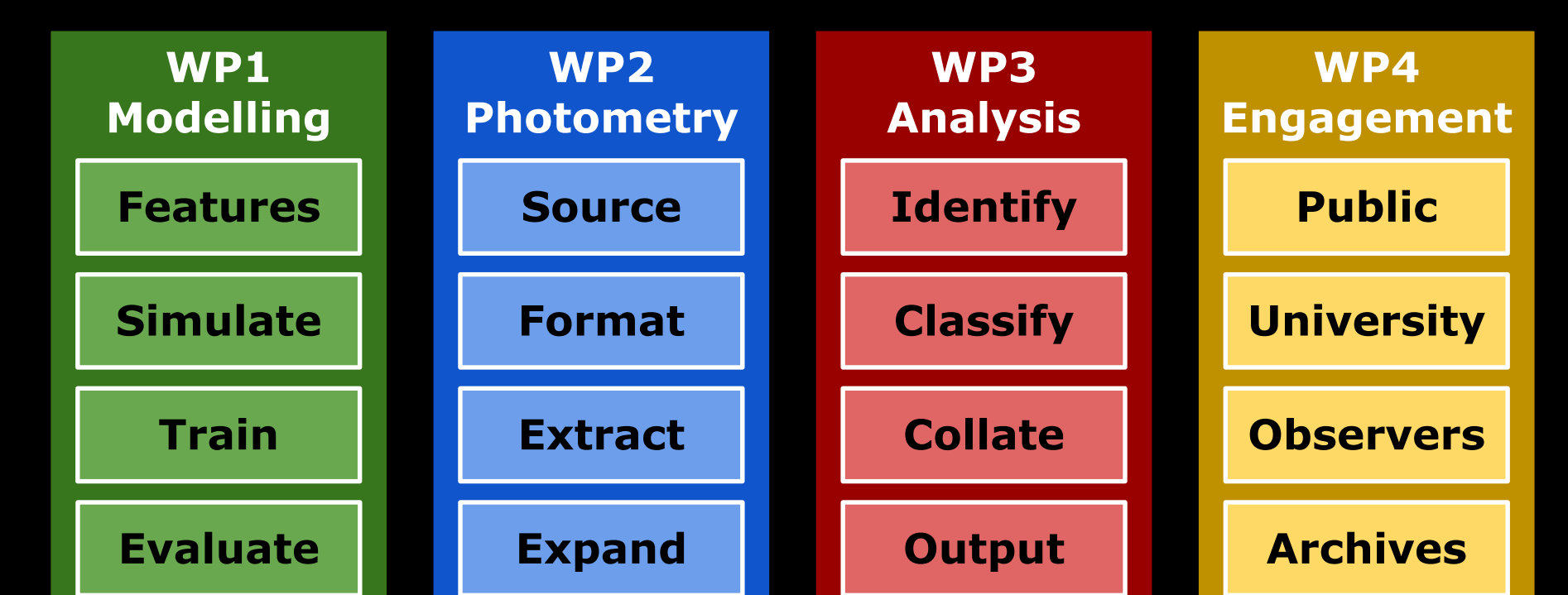
Implement simultaneous photometric processing of multiple targets in field

Assess output quality against legacy solutions

Automate reference star selection

Test impact of heterogenous data on software and database design and performance

Expansion and Development



Outline of ideal work packages, team composition, aptitudes and responsibilities

Funding Applications pending and under development: Templeton Intelligence Venture, ERC Consolidator, Research Ireland Investigator

Incremental work on individual WPs by PI and short-term research students.

Outreach and Public Engagement is an essential part of this project

Additional targets may include technosignatures or stellar variability studies